

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Dept. Name:CSE -AIDS

Course Code/Course Name: DSA06 /Data Handling and Visualization

LIST OF EXPERIMENTS

Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID Product Name January Sales February Sales

March

Sales

1 Product A 2000 2200 2400

2 Product B 1500 1800 1600

3 Product C 1200 1400 1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

3. Build a table to show the monthly sales data for each product. Label the table elements.

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

SOLUTION:

1. Grouped Bar Chart

A grouped bar chart is used to compare the sales of Product A, Product B, and Product C across January, February, and March.

First Quarter Product Sales

Grouped comparison of monthly sales for Products A, B, and C.

Product A

Product B

Product C
06501.3K2K2.6K
January
February
March

Chart elements:

- X-axis: Months
- Y-axis: Sales
- Legend: Product A, Product B, Product C
- Title: First Quarter Product Sales

2. Stacked Area Chart

The stacked area chart shows the overall sales trend and contribution of each product during the three months.

Note: A stacked area chart is normally created directly in R/Excel/Tableau; the chart below is the data structure you should use.

Monthly total sales:

- January = $2000 + 1500 + 1200 = 4700$
- February = $2200 + 1800 + 1400 = 5400$
- March = $2400 + 1600 + 1100 = 5100$

Interpretation: Overall sales increased from January to February and then decreased slightly in March.

R Code

```
# Product sales data
sales <- data.frame(
  Month = c("January", "February", "March"),
  Product_A = c(2000, 2200, 2400),
  Product_B = c(1500, 1800, 1600),
  Product_C = c(1200, 1400, 1100)
)
```

```

# Convert data to matrix
sales_matrix <- t(as.matrix(sales[, 2:4]))

# Stacked area chart
matplot(
  sales$Month,
  t(sales_matrix),
  type = "n",
  xlab = "Month",
  ylab = "Sales",
  main = "Stacked Area Chart - Product Sales"
)

polygon(
  c(1:3, 3:1),
  c(sales_matrix[1, ], rep(0, 3)),
  border = NA
)

polygon(
  c(1:3, 3:1),
  c(
    sales_matrix[1, ] + sales_matrix[2, ],
    rev(sales_matrix[1, ])
  ),
  border = NA
)

polygon(
  c(1:3, 3:1),
  c(
    sales_matrix[1, ] +
    sales_matrix[2, ] +
    sales_matrix[3, ],
    rev(sales_matrix[1, ] + sales_matrix[2, ])
  ),
  border = NA
)

```

)

```
legend(  
  "topleft",  
  legend = c("Product A", "Product B", "Product C"),  
  fill = 1:3  
)
```

3. Monthly Sales Data Table

Product ID	Product Name	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100
Total	Overall Sales	4700	5400	5100

Table elements:

- Table title: Monthly Product Sales
- Rows: Products
- Columns: Months
- Values: Monthly sales
- Total row: Overall monthly sales

4. Tableau Dashboard

Steps to create the dashboard in Tableau

Step 1 – Enter the data

Create an Excel file with:

Product ID	Product Name	January	February	March
1	Product A	2000	2200	2400

2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Open Tableau → Connect → Microsoft Excel and select the file.

Step 2 – Create Grouped Bar Chart

1. Create a new worksheet.
2. Convert January, February and March into a suitable month/value structure using Pivot.
3. Drag Month to Columns.
4. Drag Sales to Rows.
5. Drag Product Name to Color.
6. Select Bar from the Marks card.
7. Enable Show Mark Labels.
8. Rename the worksheet Grouped Bar Chart.

Step 3 – Create Stacked Area Chart

1. Create another worksheet.
2. Drag Month to Columns.
3. Drag Sales to Rows.
4. Drag Product Name to Color.
5. Change Marks type to Area.
6. Ensure the products are stacked.
7. Rename it Stacked Area Chart.

Step 4 – Create the Table

1. Create another worksheet.
2. Drag Product Name to Rows.
3. Drag Month to Columns.
4. Drag Sales to Text.
5. Enable Show Mark Labels.
6. Rename it Monthly Sales Table.

Step 5 – Create Dashboard

1. Click New Dashboard.
2. Set the dashboard size to Automatic.
3. Add:
 - Grouped Bar Chart
 - Stacked Area Chart
 - Monthly Sales Table
4. Add filters for Product Name and Month.
5. Add a suitable title:

“Product Sales Analysis – First Quarter”

6. Use dashboard actions so selecting a product in one chart filters/highlights the other views.

Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

1 28 Female 50000

2 35 Male 60000

3 42 Female 75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

SOLUTION:

1. Bar Chart – Distribution of Customer Ages

The bar chart represents the age of each customer.

Customer Age Distribution

Age of each customer in the demographic dataset.

0153045

Customer 1

Customer 2

Customer 3

Chart elements:

- X-axis: Customer
- Y-axis: Age (years)
- Title: Customer Age Distribution
- Bars: Represent the age of each customer

R Code

```
# Customer demographics data
```

```
customer <- data.frame(
```

```
  Customer_ID = c(1, 2, 3),
```

```
  Age = c(28, 35, 42),
```

```
  Gender = c("Female", "Male", "Female"),
```

```
  Income = c(50000, 60000, 75000)
```

```
)
```

```
# Bar chart

barplot(

  customer$Age,

  names.arg = paste("Customer", customer$Customer_ID),

  xlab = "Customer",

  ylab = "Age (Years)",

  main = "Customer Age Distribution",

  col = "skyblue"

)
```

2. Pie Chart – Distribution by Gender

There are:

- Female: 2 customers
- Male: 1 customer

Therefore:

- Female = 66.67%
- Male = 33.33%

R Code

```
# Gender distribution
```

```
gender_count <- table(customer$Gender)
```

```
# Pie chart
```

```
pie(

  gender_count,
```

```

main = "Customer Gender Distribution",

labels = paste(

  names(gender_count),

  round(100 * gender_count / sum(gender_count), 2),

  "%")

)

)

```

Customer Gender Distribution

Distribution of customers by gender.

Female

Male

Interpretation: Female customers make up 66.67%, while male customers make up 33.33% of the dataset.

3. Customer Demographic Information Table

Customer Demographics

Customer ID	Age	Gender	Income (\$)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

Table elements:

- Table title: Customer Demographics
- Rows: Individual customers
- Columns: Customer ID, Age, Gender, Income
- Income: Shown in US dollars

R Code

```
# Display customer demographic table
```

```
print(customer)
```

4. Tableau Dashboard

The Tableau dashboard combines the bar chart, pie chart, and demographic table for interactive analysis.

Step 1 – Enter the Data

Create an Excel file:

Customer ID	Age	Gender	Income
-------------	-----	--------	--------

1	28	Female	50000
---	----	--------	-------

2	35	Male	60000
---	----	------	-------

3	42	Female	75000
---	----	--------	-------

Open:

Tableau → Connect → Microsoft Excel → Select the dataset

Step 2 – Create the Age Bar Chart

1. Create a new worksheet.
2. Drag Customer ID to Columns.
3. Drag Age to Rows.
4. Select Bar from the Marks card.
5. Turn on Show Mark Labels.
6. Add the title Customer Age Distribution.

Step 3 – Create the Gender Pie Chart

1. Create a new worksheet.
2. Select Pie from the Marks card.
3. Drag Gender to Color.
4. Drag Customer ID to Angle.
5. Drag Customer ID to Label.
6. Add the title Customer Gender Distribution.

Step 4 – Create the Demographic Table

1. Create a new worksheet.
2. Drag Customer ID to Rows.
3. Drag Age, Gender, and Income to the appropriate table/text fields.
4. Format Income as currency.
5. Add the title Customer Demographic Information.

Step 5 – Create the Dashboard

1. Click New Dashboard.
2. Add the following three worksheets:
 - Customer Age Distribution
 - Customer Gender Distribution
 - Customer Demographic Information
3. Arrange them neatly on the dashboard.
4. Add filters for:
 - Gender
 - Age
 - Income

Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

Employee ID Department Years of Service Performance Score

2 HR 3 92

3 Marketing 7 78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and

title the chart.

2. Generate a bar chart showing the distribution of employees across different departments. Label the

chart elements.

3. Build a table to display the performance data for each employee. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive

exploration of employee performance data.

SOLUTION:

1. Line Chart – Performance Trend

The line chart shows the performance score according to years of service.

Employee Performance Trend

Performance score plotted against years of service.

7580859095357

Chart elements:

- X-axis: Years of Service
- Y-axis: Performance Score
- Title: Employee Performance Trend
- Data points: Employee performance scores

R Code

```
# Employee performance data
```

```
employee <- data.frame(
```

```
  Employee_ID = c(1, 2, 3),
```

```
  Department = c("Sales", "HR", "Marketing"),
```

```
  Years_of_Service = c(5, 3, 7),
```

```
  Performance_Score = c(85, 92, 78)
```

```
)
```

```
# Sort data according to years of service
```

```
employee <- employee[order(employee$Years_of_Service), ]
```

```
# Line chart
```

```
plot(
```

```
  employee$Years_of_Service,
```

```
  employee$Performance_Score,
```

```
  type = "o",
```

```
  xlab = "Years of Service",
```

```

ylab = "Performance Score",

main = "Employee Performance Trend"

)

```

Interpretation: The highest performance score is 92 for the employee with 3 years of service. The score decreases to 85 at 5 years and 78 at 7 years.

2. Bar Chart – Employees by Department

Each department has one employee.

Department	Number of Employees
Sales	1
HR	1
Marketing	1

Employees by Department

Number of employees in each department.

012

Sales

HR

Marketing

R Code

```
# Count employees by department
```

```
dept_count <- table(employee$Department)
```

```
# Bar chart
```

```
barplot(
```

```
  dept_count,
```

```
  xlab = "Department",
```

```
  ylab = "Number of Employees",
```

```
  main = "Employees by Department"
```

```
)
```

Chart elements:

- X-axis: Department
- Y-axis: Number of Employees
- Title: Employees by Department
- Bars: Sales, HR, and Marketing

3. Employee Performance Table

Employee Performance Data

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Table elements:

- Table title: Employee Performance Data
- Rows: Individual employees
- Columns: Employee ID, Department, Years of Service, Performance Score
- Values: Employee performance information

R Code

```
# Display employee performance table
```

```
print(employee)
```

4. Tableau Dashboard

Step 1 – Enter the Data

Create an Excel file containing:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Open:

Tableau → Connect → Microsoft Excel → Select the file

Step 2 – Create the Line Chart

1. Create a new worksheet.
2. Drag Years of Service to Columns.
3. Drag Performance Score to Rows.
4. Select Line from the Marks card.
5. Drag Employee ID to Label.
6. Turn on Show Mark Labels.
7. Give the worksheet the title Employee Performance Trend.

Step 3 – Create the Department Bar Chart

1. Create a new worksheet.
2. Drag Department to Columns.
3. Drag Employee ID to Rows.

4. Change the aggregation to Count.
5. Select Bar from the Marks card.
6. Enable Show Mark Labels.
7. Title it Employees by Department.

Step 4 – Create the Performance Table

1. Create a new worksheet.
2. Drag Employee ID to Rows.
3. Drag Department to Rows.
4. Drag Years of Service to Rows.
5. Drag Performance Score to Rows/Text.
6. Format it as a table.
7. Title it Employee Performance Data.

Step 5 – Create the Tableau Dashboard

1. Click New Dashboard.
2. Add the three worksheets:
 - Employee Performance Trend
 - Employees by Department
 - Employee Performance Data
3. Arrange the charts and table neatly.
4. Add filters for:
 - Department
 - Years of Service
 - Employee ID
5. Enable dashboard highlighting/filtering so selecting an employee or department updates the other views.

SET 9

Dataset: Online_Learning_Activity.csv

Student_ID Gender Age Course Study_Time (hrs) Videos_Watched Quiz_Score
Login_Date

L01 Male 20 R 3.5 12 78 2025-01-05

L02 Female 22 R 4.2 15 85 2025-01-05

L03 Male 19 SQL 2.0 8 65 2025-02-08

L04 Female 21 R 5.0 18 92 2025-02-08

L05 Male 23 R 2.5 9 70 2025-03-12

L06 Female 20 SQL 4.0 14 88 2025-03-12

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course.

Add title and labels. Interpret student performance.

2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per

month. Plot a line chart. Apply moving average smoothing. Identify the trend.

4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score

by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course.

Design a mini interactive dashboard.

SOLUTION:

1. Histogram and Boxplot

R Code

```
# Import the dataset
```

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
# View the data
```

```
print(data)
```

```
# Histogram of Quiz Score
```

```
hist(
```

```
  data$Quiz_Score,
```

```
  main = "Distribution of Quiz Scores",
```

```
  xlab = "Quiz Score",
```

```
  ylab = "Number of Students",
```

```
  col = "skyblue",
```

```
  border = "black"
```

```
)
```

```
# Boxplot of Quiz Score by Course
```

```
boxplot(
```

```
  Quiz_Score ~ Course,
```

```
  data = data,
```

```
  main = "Quiz Score by Course",
```

```
  xlab = "Course",
```

```
ylab = "Quiz Score",  
  
col = "lightgreen"  
  
)
```

Interpretation

- Quiz scores range from 65 to 92.
- The highest score is 92, while the lowest is 65.
- Students studying R have scores of 78, 85, 92 and 70.
- Students studying SQL have scores of 65 and 88.
- The R course has more students and therefore provides a better representation of the overall score distribution.

2. Scatter Plot – Study Time vs Quiz Score

The scatter plot shows the relationship between the amount of time students study and their quiz scores.

Videos_Watched is represented using bubble size, while transparency helps overlapping observations remain visible.

R Code

```
# Scatter plot
```

```
plot(  
  
  data$Study_Time,  
  
  data$Quiz_Score,  
  
  pch = 21,  
  
  bg = rgb(0.2, 0.5, 0.8, 0.5),  
  
  cex = data$Videos_Watched / 4,  
  
  xlab = "Study Time (Hours)",
```

```
ylab = "Quiz Score",  
  
main = "Study Time vs Quiz Score"  
  
)  
  
# Add student labels  
  
text(  
  
  data$Study_Time,  
  
  data$Quiz_Score,  
  
  labels = data$Student_ID,  
  
  pos = 3  
  
)
```

Interpretation

There is a generally positive learning pattern:

- L03 studied 2 hours and scored 65.
- L05 studied 2.5 hours and scored 70.
- L01 studied 3.5 hours and scored 78.
- L06 studied 4 hours and scored 88.
- L02 studied 4.2 hours and scored 85.
- L04 studied 5 hours and scored 92.

Overall, students who spent more time studying generally achieved higher quiz scores. Students who watched more videos also tended to have higher scores, although the relationship is not perfectly linear.

3. Monthly Average Quiz Score and Moving Average

First, convert `Login_Date` into Date format and extract the month.

R Code

```
# Convert Login_Date into Date format
data$Login_Date <- as.Date(data$Login_Date)

# Extract month
data$Month <- format(data$Login_Date, "%Y-%m")

# Calculate average quiz score per month
monthly_avg <- aggregate(
  Quiz_Score ~ Month,
  data = data,
  FUN = mean
)

# Display monthly averages
print(monthly_avg)
```

Monthly Average Results

Month	Average Score	Quiz
-------	------------------	------

2025-01	81.5	
---------	------	--

2025-02	78.5	
---------	------	--

2025-03	79.0	
---------	------	--

Moving Average

Because there are only three months, a 2-month moving average is suitable.

Calculate 2-month moving average

```
monthly_avg$Moving_Average <- c(
  NA,
  (monthly_avg$Quiz_Score[1] +
   monthly_avg$Quiz_Score[2]) / 2,
  (monthly_avg$Quiz_Score[2] +
   monthly_avg$Quiz_Score[3]) / 2
)
```

```
# Display results
```

```
print(monthly_avg)
```

Moving Average Results

Month	Average Score	Quiz	2-Month Average	Moving
2025-01	81.5		NA	
2025-02	78.5		80.0	
2025-03	79.0		78.75	

Line Chart with Moving Average

```
plot(  
  monthly_avg$Month,  
  monthly_avg$Quiz_Score,  
  type = "o",  
  xlab = "Month",  
  ylab = "Average Quiz Score",  
  main = "Monthly Average Quiz Score",  
  ylim = c(75, 85)
```

```

)
lines(
  monthly_avg$Month,
  monthly_avg$Moving_Average,
  type = "o",
  lty = 2
)
legend(
  "topright",
  legend = c("Average Score", "2-Month Moving Average"),
  lty = c(1, 2)
)

```

Trend Interpretation

- January has the highest monthly average: 81.5.
- The average decreases to 78.5 in February.
- It increased slightly to 79.0 in March.
- Therefore, the overall trend shows a slight decrease followed by a small recovery.

4. Tableau Mini Interactive Dashboard

Step 1 – Import Dataset

Open Tableau:

Connect → Text File → Online_Learning_Activity.csv

Make sure:

- Study_Time → Number
- Videos_Watched → Number
- Quiz_Score → Number
- Age → Number
- Login_Date → Date

Step 2 – Bar Chart: Average Quiz Score by Course

Create a new worksheet.

1. Drag Course → Columns.
2. Drag Quiz_Score → Rows.
3. Change aggregation to Average.
4. Select Bar Chart.
5. Enable Show Mark Labels.
6. Title it:

Average Quiz Score by Course

The averages are:

Course	Average Score	Quiz
--------	------------------	------

R	81.25	
---	-------	--

SQL	76.50	
-----	-------	--

Step 3 – Scatter Plot

Create another worksheet.

1. Drag Study_Time → Columns.
2. Drag Quiz_Score → Rows.
3. Select Circle from Marks.

4. Drag Videos_Watched → Size.
5. Drag Student_ID → Detail.
6. Drag Gender → Color.
7. Add a trend line if required.

Chart Title

Study Time vs Quiz Score

This allows you to see whether increased study time is associated with better quiz performance.

Step 4 – Add Gender and Course Filters

For the scatter plot:

1. Drag Gender → Filters.
2. Drag Course → Filters.
3. Right-click each filter.
4. Select Show Filter.

Now users can select:

- Male/Female
- R/SQL
- Both or individual categories.

Step 5 – Create Mini Dashboard

Click New Dashboard.

Add:

1. Average Quiz Score by Course

Bar chart at the top.

2. Study Time vs Quiz Score

Scatter plot below the bar chart.

3. Filters

Place Gender and Course on the side.

Dashboard Title

Online Learning Activity Analysis

Dashboard Features

- Compare average performance between R and SQL.
- Analyze the relationship between study time and quiz score.
- See the effect of videos watched through bubble size.
- Filter results by Gender.
- Filter results by Course.
- Explore individual student performance interactively.